Cheng Xi

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Education

University of California, Santa Barbara — Santa Barbara, CA

Bachelor of Science in Computer Science — *Sep. 2023 – June 2027*

Relevant Coursework: Data Structures & Algorithms, Object Oriented Design, Advanced App Programming, Graph & Network Theory, Computer Architecture, Artificial Intelligence, Operating Systems, Distributed Systems, Deep Learning, Machine Learning, Parallel Programming, Advanced Data Structures & Algorithms

Experience

Systems Engineering Intern — June 2026 – September 2026

Woven by Toyota — *Ann Arbor, Michigan*

- Collaborated with software developers, software security team, and design team to create comprehensive development and security solutions for software-defined vehicles
- Researched and developed system designs, architecture, requirements, test cases using *CATIA* and *Sysml2* for a robust and secure Source Development Kit.
- Developed multi-threaded prototypes in *C/C++, Rust*, enabling simulated proof of concept demonstrations in realistic and performance/safety critical environments.

Systems Engineering Intern — January 2026 – June 2026

Minno — *Santa Barbara, CA*

- Developed *cloud storage* and *distributed network* solutions using *Google Cloud*, ensured *0%* downtime for active services across all connected *OTA* devices.
- Developed *AI*, *MDM* and *OTA* solutions for enterprise companies, including on-device agents using small language models - *SLMs* and machine vision based tools.
- Developed automated *Bash* and *Powershell* diagnostic scripts to extract device specifications (RAM, CPU, storage, network modules) and log critical hardware/driver health status, reducing diagnostic time by 70%.

Software Engineer - Ground Support Lead — Sep. 2024 – Present

Gaucho Rocket Program — *Santa Barbara, CA*

- Leading a team of 7 software engineers and 3 hardware engineers to design, code, and build the first open-source ground control station for rocketry, with the codebase deriving from *QGroundControl*. Utilizes *CI/CD* via *Github Actions* to ensure seamless deployment across multiple platforms.
- Programmed and tested code for flight data calculation and reaction wheel control in *C/C++*, enabled backwards compatibility crucial for flight computer software which resulted in zero exceptions thrown.
- Built and tested multi-threaded modules for the rocket flight computer, allowing for more precise flight data calculations and achieving greater than 50% latency reduction in height and Euler angle calculations.
- Developed and debugged thrust vector control module, working in tandem with the electronics/design branch to account for servo offset. Created a module resulting in angular accuracy within 0.1 degrees of error.

Undergraduate Researcher — Jun. 2025 – April 2026

Elizabeth M. Belding MOMENT Lab — *Santa Barbara, CA*

- Developed and tested computer network solutions using *L4S*, responsible for integrating and testing key *SCReAM* congestion control modules such as deque based queueing systems for packets. Reducing delay by 60%
- Found and fixed critical errors preventing proper communication between the sender and receiver component of *SCReAM*, a breakthrough which allowed for video transmission rates of more than 15,000kbps from less than 2,000kbps previously.
- Created data analytics tools using *Python* libraries *pandas*, *numpy re(Regex)*, and *Matplotlib*, which automatically analyzed and visualized over 500,000 lines of data collected from experiments.
- Co-authored research paper utilizing *L4S* in video conferencing / streaming applications, published on *Arxiv*

Software Engineer - AI Consultant — Feb. 2024 – Oct. 2024

Scale AI — *Remote*

- Prompted and analyzed LLM outputs for hallucination and error, recorded instances of incorrect information and provided correction.
- Built testbed suite in *Python*, allowed for automatic cross-referencing of information from LLM outputs to *Google* searches.

Projects

droneCV | *C, C++, Git, TypeScript, CMake* Dec. 2024 – Present

- Developed a real-time object tracking system for drones using *C++* and *OpenCV*, implementing and evaluating *DIS*, *DualTVL1O*, *Farneback*, and *DenseRLOF* optical flow algorithms.
- Utilized multi-threading techniques and parallel programming leveraging GPUs for real-time object tracking, resulting in an increase from 30FPS to 60FPS in a 1024x768 resolution camera.

LeveragedPrediction | *Python* Jan. 2025 – June 2026

- Developed a machine learning model to predict leveraged ETF prices using *Python (Scikit-learn, TensorFlow)*; combined quantitative time-series analysis with *SEC* fundamental data.
- Implemented a gradient boosting decision tree to analyze correlation between VIX and increased leveraged ETF volatility, allowing for accurate predictions given the VIX price. Implemented a recurrent neural network to encapsulate and train a predictive model, achieving approximately 95% accuracy in historical data.

CTAPIDDF | *Python, PostgreSQL* Oct. 2024 – Jan. 2025

- Developed a script to gather *FDA* trial data utilizing *Python*. Utilized resulting data to invest in biotechnology companies, with aggregated returns of 91.5% from November 2024 - October 2025.
- Implemented a *SQL* database to categorize trials based attributes and sponsors, allowing for categorization of approximately 19,000 active clinical trials.

Technical Skills

Languages: C/C++, CUDA, Rust, Java, Python, SQL (Postgres), JavaScript, HTML/CSS, Swift
Frameworks: React, Node.js, FastAPI, Spring Boot, Prisma, Angular
Developer Tools: Git, Docker, Kubernetes, Google Cloud Platform, VS Code, Visual Studio, Qt Tools, Qt Creator, IntelliJ, Eclipse, Maven, Dokku, NPM, Powershell, Bash, CI/CD, CATIA Magic, CATIA Collaborator, JAMA, Jira, Aha
Libraries: pandas, NumPy, re, Matplotlib, OpenCV, TensorFlow, Pytorch, polars, Scikit-learn, Syside, Sysml2.0, UML, Sysml